

# Completely Bounded Schur Multipliers Between Mixed Schatten Classes

A full candidate solution to Problem 7.7 of arXiv:math/0505306

Automated mathematics run `fa_banach_001`, lane 6

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**Status.** Full solution candidate, likely valid, awaiting specialist review. The main review point is the Hilbertian torus average in Lemma 4: it is performed in  $S_2^m[S_2^n] = S_2^{mn}$  and then converted back to complete boundedness by the  $S_2$ -amplification criterion.

## 1 Source problem

The source is Q. Xu, *Operator-space Grothendieck inequalities for noncommutative  $L_p$ -spaces*, arXiv:math/0505306, later published in *Duke Mathematical Journal* **131** (2006), 525–574 [1]. For a scalar matrix  $\varphi = (\varphi_{ij})$ , its Schur multiplier is

$$M_\varphi(x) = (\varphi_{ij}x_{ij}).$$

The Schatten classes carry their canonical operator-space structures. Xu characterizes bounded multipliers  $S_p \rightarrow S_q$  throughout  $1 \leq q \leq 2 \leq p \leq \infty$ , and completely bounded multipliers on the dual line  $S_p \rightarrow S_{p'}$ . Problem 7.7 asks for the completely bounded characterization away from that line.

for  $i'$ ), we get a Schur extension of  $\varphi$ . □

The second part of Corollary 7.6 gives a characterization of c.b. Schur multipliers from  $S_p$  to  $S_{p'}$ , i.e. from  $S_p$  to its dual. We do not know how to characterize the c.b. Schur multipliers from  $S_p$  into  $S_q$  for any  $1 \leq q \leq 2 \leq p \leq \infty$ , as in the first part at the Banach space level.

**Problem 7.7.** Let  $1 \leq q \leq 2 \leq p \leq \infty$  with  $p \neq q'$ . Characterize c.b. Schur multipliers from  $S_p$  into  $S_q$ , in a way similar to that in Corollary 7.6

This problem might be related to the following

**Problem 7.8.** Let  $M$  be a von Neumann algebra and  $p, q \geq 2$  with  $p \neq q$ . Let  $E \subset L_p(M)$  and  $F \subset L_q(M)$ . Find a Grothendieck type inequality for j.c.b. forms  $u : E \times F \rightarrow \mathbb{C}$ .

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Figure 1: Problem 7.7 and the adjacent mixed Grothendieck problem in the source, printed page 28 (PDF page 29).

## 2 Statement and proof idea

Put

$$\frac{1}{r} = \frac{1}{q} - \frac{1}{p}, \quad 1 \leq q \leq 2 \leq p \leq \infty, \quad (1)$$

with the usual convention  $r = \infty$  if  $p = q = 2$ . For  $1 \leq s \leq \infty$ , define the positive domination norm

$$\mathfrak{F}_s(\varphi) = \inf \{ \|a\|_s \|b\|_s : |\varphi_{ij}| \leq a_i b_j, \ a, b \geq 0 \}. \quad (2)$$

**Theorem 1** (Mixed completely bounded Schur multipliers). *Let  $1 \leq q \leq 2 \leq p \leq \infty$ , and let  $r$  be given by (1). A scalar matrix  $\varphi$  defines a completely bounded Schur multiplier*

$$M_\varphi : S_p \longrightarrow S_q$$

*if and only if  $\mathfrak{F}_{2r}(\varphi) < \infty$ . Quantitatively, there is a constant  $C_{p,q} < \infty$ , depending only on  $p, q$ , such that*

$$C_{p,q}^{-1} \mathfrak{F}_{2r}(\varphi) \leq \|M_\varphi : S_p \rightarrow S_q\|_{\text{cb}} \leq \mathfrak{F}_{2r}(\varphi). \quad (3)$$

*For  $p = q = 2$ , this says simply that  $\varphi$  is bounded.*

The central space is  $S_2$ , the homogeneous operator Hilbert space. The easy direction splits an  $\ell_{2r}$  weight into the portions needed for  $S_p \rightarrow S_2$  and  $S_2 \rightarrow S_q$ . For necessity, operator-space Maurey factorization first extracts the  $S_p \rightarrow S_2$  weights. Schur symmetry lets us choose them diagonal. The residual is a cb Schur map  $S_2 \rightarrow S_q$ ; the operator-space little Grothendieck theorem extracts its endpoint weights. Hölder's identity combines the two pairs at exactly the exponent  $2r$ .

## 3 Three ingredients

Write

$$\delta_1 = \frac{1}{2} - \frac{1}{p}, \quad \delta_2 = \frac{1}{q} - \frac{1}{2}, \quad \delta = \delta_1 + \delta_2 = \frac{1}{r}. \quad (4)$$

**Lemma 2** (Diagonal multiplication and the Hilbert midpoint). *Assume  $\delta > 0$ . If  $a, b \in \ell_{2r}$  are nonnegative and  $|\varphi_{ij}| \leq a_i b_j$ , then*

$$\|M_\varphi : S_p \rightarrow S_q\|_{\text{cb}} \leq \|a\|_{2r} \|b\|_{2r}.$$

*Proof.* Let  $\theta_k = \delta_k / \delta$ , so  $\theta_1 + \theta_2 = 1$ , and put  $\psi_{ij} = \varphi_{ij} / (a_i b_j)$  when  $a_i b_j \neq 0$ , and  $\psi_{ij} = 0$  otherwise. Then  $\|\psi\|_\infty \leq 1$ . The diagonal multiplication maps

$$J_1(x) = \text{diag}(a^{\theta_1}) x \text{diag}(b^{\theta_1}) : S_p \rightarrow S_2,$$

and

$$J_2(y) = \text{diag}(a^{\theta_2}) y \text{diag}(b^{\theta_2}) : S_2 \rightarrow S_q$$

are completely bounded by noncommutative Hölder at every matrix level. Indeed, when  $\delta_k > 0$ , the corresponding weight exponent is

$$\frac{2r}{\theta_k} = \frac{2}{\delta_k};$$

when  $\delta_k = 0$ , the associated map is omitted. Moreover,

$$\|J_k\|_{\text{cb}} \leq \|a\|_{2r}^{\theta_k} \|b\|_{2r}^{\theta_k}.$$

Since  $S_2$  is a homogeneous operator Hilbert space,  $\|M_\psi : S_2 \rightarrow S_2\|_{\text{cb}} = \|\psi\|_\infty \leq 1$ . Finally

$$M_\varphi = J_2 M_\psi J_1,$$

which gives the assertion.  $\square$

**Lemma 3** (Little Grothendieck endpoint). *Let  $2 < s \leq \infty$ . If  $M_\eta : S_s \rightarrow S_2$  is completely bounded, then there are nonnegative  $c, d \in \ell_{4s/(s-2)}$  such that*

$$|\eta_{ij}| \leq c_i d_j, \quad \|c\|_{4s/(s-2)} \|d\|_{4s/(s-2)} \leq K_s \|M_\eta\|_{\text{cb}}.$$

*The transposed dual statement holds for  $M_\eta : S_2 \rightarrow S_{s'}$ .*

*Proof.* Apply Xu's operator-space little Grothendieck theorem [1, Corollary 7.2] to the map  $S_s \rightarrow S_2 = OH$ . It supplies positive unit functionals  $f, g$  on  $S_{s/2}$  such that

$$\|M_\eta(x)\|_2 \leq K_s \|M_\eta\|_{\text{cb}} f(x^*x)^{1/4} g(xx^*)^{1/4}.$$

Testing  $x = e_{ij}$  yields

$$|\eta_{ij}| \leq K_s \|M_\eta\|_{\text{cb}} f(e_{jj})^{1/4} g(e_{ii})^{1/4}.$$

The diagonal of a positive unit element of  $(S_{s/2})^* = S_{s/(s-2)}$  has norm at most one in  $\ell_{s/(s-2)}$ . Its fourth root therefore belongs to  $\ell_{4s/(s-2)}$ . For  $s = \infty$ , a state may have a singular part, but the sequence of its values on the minimal diagonal projections still lies in the unit ball of  $\ell_1$ . This proves the first assertion. The second follows by taking adjoints.  $\square$

The next lemma is the only place where mixed exponents enter.

**Lemma 4** (Equivariant Maurey factorization). *Let  $1 \leq q < 2 < p \leq \infty$ , initially in  $n$  dimensions. If  $T = M_\varphi : S_p^n \rightarrow S_q^n$  is completely bounded, then there are nonnegative diagonal weights  $a, b \in \ell_{2/\delta_1}^n$  and a completely bounded Schur multiplier  $M_\psi : S_2^n \rightarrow S_q^n$  such that*

$$\varphi_{ij} = a_i \psi_{ij} b_j \tag{5}$$

and

$$\|a\|_{2/\delta_1} \|M_\psi : S_2^n \rightarrow S_q^n\|_{\text{cb}} \|b\|_{2/\delta_1} \leq A_{p,q} \|T\|_{\text{cb}}, \tag{6}$$

where  $A_{p,q}$  is independent of  $n$ .

*Proof.* Junge–Parcet's Lemma 4.1(iii) says that a cb map into  $S_q$  is completely  $(q, 1)$ -summing with summing norm at most its cb norm. Apply their operator-space Maurey theorem [2, Theorem A(iii)] with domain exponent  $p$ , summing exponent  $q$ , and intermediate exponent 2. It gives a factorization

$$T(x) = V(d_1 x d_2), \quad V : S_2^n \rightarrow S_q^n,$$

where  $d_1, d_2 \in S_{2/\delta_1}^n$  are positive, with the product of the three norms bounded as in (6). If  $p = \infty$ , use Theorem A(i) instead; it gives  $d_1, d_2 \in S_4^n$ , again  $4 = 2/\delta_1$ .

Let  $E_\Delta : M_n \rightarrow \ell_\infty^n$  be the diagonal conditional expectation and set

$$a = E_\Delta(d_1^2)^{1/2}, \quad b = E_\Delta(d_2^2)^{1/2}.$$

These are diagonal, and conditional-expectation contractivity on  $S_{1/\delta_1}^n$  gives

$$\|a\|_{2/\delta_1} \leq \|d_1\|_{2/\delta_1}, \quad \|b\|_{2/\delta_1} \leq \|d_2\|_{2/\delta_1}.$$

We claim that  $T$  factors completely boundedly through  $x \mapsto axb$ . Let  $m \geq 1$  and  $x \in S_2^m[S_p^n]$ . For diagonal unitaries  $U, W$ , Schur equivariance and unitary invariance give

$$\|(I_{S_2^m} \otimes T)x\|_{S_2^m[S_q^n]} = \|(I_{S_2^m} \otimes T)(UxW)\|_{S_2^m[S_q^n]}.$$

Use  $T(z) = V(d_1 z d_2)$ , square, and integrate over the two independent diagonal tori. Since  $S_2^m[S_2^n] = S_2^{mn}$  is Hilbertian, Fubini and orthogonality of the characters give the exact identity

$$\begin{aligned} \|(I_{S_2^m} \otimes T)x\|_{S_2^m[S_q^n]}^2 &\leq \|V\|_{\text{cb}}^2 \iint \|d_1 U x W d_2\|_{S_2^{mn}}^2 dU dW \\ &= \|V\|_{\text{cb}}^2 \|axb\|_{S_2^{mn}}^2. \end{aligned}$$

Thus  $V_0(axb) = T(x)$  is well defined on the range of  $x \mapsto axb$ , and Xu's  $S_2$ -amplification criterion [1, Lemma 1.1] gives  $\|V_0\|_{\text{cb}} \leq \|V\|_{\text{cb}}$ . The range is the span of those matrix units  $e_{ij}$  for which  $a_i b_j > 0$ . Its orthogonal coordinate projection in  $S_2^n = OH_{n^2}$  is a complete contraction, by homogeneity of  $OH$ . Extending  $V_0$  by zero on the orthogonal complement therefore gives a cb map on all of  $S_2^n$ .

It remains to make this residual map Schur. Average it by

$$\tilde{V}(y) = \int_{\mathbb{T}^n \times \mathbb{T}^n} U^* V(UyW) W^* dU dW.$$

This is a cb Schur multiplier and  $\|\tilde{V}\|_{\text{cb}} \leq \|V\|_{\text{cb}}$ . Since  $a, b$  are diagonal,

$$\tilde{V}(axb) = T(x)$$

for every  $x$ . Writing  $\tilde{V} = M_\psi$  gives (5)–(6). If  $a_i b_j = 0$ , the factorization forces  $\varphi_{ij} = 0$ , so we set  $\psi_{ij} = 0$  there.  $\square$

## 4 Proof of Theorem 1

The upper bound in (3) is Lemma 2. We prove necessity.

First suppose  $1 \leq q < 2 < p \leq \infty$  and work in  $n$  dimensions. Apply Lemma 4. Thus

$$\varphi_{ij} = a_i \psi_{ij} b_j, \quad a, b \in \ell_{s_p}^n, \quad s_p = \frac{2}{\delta_1}.$$

The adjoint of  $M_\psi : S_2^n \rightarrow S_q^n$  is a Schur multiplier  $S_{q'}^n \rightarrow S_2^n$ . Lemma 3, with  $s = q'$ , gives nonnegative  $c, d \in \ell_{s_q}^n$  such that

$$|\psi_{ij}| \leq c_i d_j, \quad s_q = \frac{4q}{2-q} = \frac{2}{\delta_2}.$$

Consequently

$$|\varphi_{ij}| \leq (a_i c_i)(b_j d_j).$$

By Hölder and (4),

$$\frac{1}{2r} = \frac{\delta}{2} = \frac{\delta_1}{2} + \frac{\delta_2}{2} = \frac{1}{s_p} + \frac{1}{s_q},$$

so  $ac, bd \in \ell_{2r}^n$ . The norm bounds in the two lemmas give

$$\mathfrak{F}_{2r}(\varphi|_{[n]^2}) \leq C_{p,q} \|M_\varphi : S_p^n \rightarrow S_q^n\|_{\text{cb}},$$

with  $C_{p,q}$  independent of  $n$ .

If  $q = 2 < p$ , apply Lemma 3 directly to  $M_\varphi : S_p \rightarrow S_2$ ; its exponent is

$$\frac{4p}{p-2} = \frac{2}{\delta_1} = 2r.$$

If  $p = 2 > q$ , apply the adjoint version. If  $p = q = 2$ , homogeneity of  $S_2$  gives the claim with  $a_i = b_j = \|\varphi\|_\infty^{1/2}$ . These arguments also include  $p = \infty$  and  $q = 1$ .

Finally let  $\varphi$  be infinite. Every finite northwest compression has cb norm at most  $\|M_\varphi\|_{\text{cb}}$  and hence admits factors  $A^{(n)}, B^{(n)} \in \ell_{2r}^n$  with uniformly bounded product. Balance the two norms by scalar rescaling and extend the sequences by zero. Since  $2r \geq 2$ , bounded sets in  $\ell_{2r}$  are weakly sequentially compact. Pass to a common weakly convergent subsequence. Coordinates converge, so the pointwise domination passes to the limit, while weak lower semicontinuity preserves the norm bound. Thus  $\mathfrak{F}_{2r}(\varphi) \leq C_{p,q} \|M_\varphi\|_{\text{cb}}$ , completing the proof.

## 5 Checks, sharpness, and limitations

On the dual line  $q = p'$ , one has

$$2r = \frac{2p}{p-2},$$

so Theorem 1 is exactly Xu's Corollary 7.6(ii). At the two off-dual Hilbert corners it gives  $\ell_4 \times \ell_4$  for  $S_\infty \rightarrow S_2$  and  $S_2 \rightarrow S_1$ . At the extreme corner  $S_\infty \rightarrow S_1$  it gives  $\ell_2 \times \ell_2$ .

For rank-one symbols  $\varphi_{ij} = \alpha_i \beta_j$ , the multiplier is  $x \mapsto \text{diag}(\alpha)x \text{diag}(\beta)$ . Oikhberg proved the exact formula [3, Theorem 2.1]

$$\|x \mapsto Ax B : S_p \rightarrow S_q\|_{\text{cb}} = \|A\|_{2r} \|B\|_{2r}.$$

Thus the exponent  $2r$  is forced already by rank-one symbols. Oikhberg also records the  $S_\infty \rightarrow S_2$  Schur endpoint in his Theorems 5.1–5.2.

No numerical calculation enters the proof. The accompanying script checks the exponent identities exactly on representative rational cases.

The result is a characterization of Schur multipliers, not a solution to the broader mixed j.c.b. Problem 7.8. The Schur symmetry is used decisively to diagonalize the Maurey densities and the residual map.

## 6 Novelty check and review recommendation

The bounded search performed on 11 August 2026 covered the run registry and source indexes, exact web/arXiv phrases for mixed cb Schur multipliers and factorization through  $S_2/OH$ , and the 33 works indexed by OpenAlex as citing Xu's paper. It found the two ingredients used here—Junge–Parcet and Oikhberg—and later same-exponent work, but no paper stating the full mixed characterization. Because the proof combines known machinery in a short new way, novelty confidence is moderate rather than definitive.

Human review should first inspect Lemma 4, specifically the equality obtained by averaging the squared Hilbert–Schmidt norm and the subsequent use of the  $S_2$ -amplification criterion. Once this diagonal factorization is accepted, the remaining argument is the cited endpoint theorem plus Hölder arithmetic.

## References

- [1] Q. Xu, *Operator-space Grothendieck inequalities for noncommutative  $L_p$ -spaces*, Duke Math. J. **131** (2006), 525–574; arXiv:math/0505306.
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- [3] T. Oikhberg, *Completely bounded and ideal norms of multiplication operators and Schur multipliers*, Integral Equations Operator Theory **66** (2010), 425–440, doi:10.1007/s00020-010-1751-5.